Independent projectile directions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why horizontal and vertical projectile motion share time but evolve differently.

SUCCESS CRITERIA - I can...

1. Identify horizontal and vertical velocity components.
2. Use the same elapsed time for both axes.
3. State the negligible-air-resistance assumption.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “projectile” and “horizontal” to explain your model or result. Sentence starters are optional.

projectile: an object moving after launch under the stated forces

horizontal: along a level line, usually the x direction

vertical: up and down, usually the y direction

01 Bring back an earlier idea

A 3 kg test mass feels 15 N toward a source.
Find g and the force on a 7 kg mass at the same point.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $a_x = 0$; $a_y = -g$; v_x = constant.
Upward positive; g = 10 m/s². Ideal near-Earth motion with negligible air resistance.

Worked reasoning: A ball launches horizontally at 3 m/s and falls for 2 s. Horizontal travel is 3×2=6 m. With g=10 m/s² and no initial vertical velocity, its final vertical velocity is 20 m/s down.

Your turn: Draw separate horizontal and vertical velocity arrows at launch and after 2 s.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Look and explain

One ball is dropped and another launched horizontally from the same height at the same instant.
Compare fall times and explain.

Sentence starter: Both ... . They differ in ... because ... .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

04 Build the solution

A horizontal launch has v_x=3 m/s and stays in air 2 s.
Find horizontal travel. What happens ifv_x doubles but height stays the same?

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

A student says a larger horizontal speed holds the ball up longer.
Explain which force or acceleration would be needed for that claim.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Choose a method

A student says a 4 kg cart under 24 N net force accelerates at 96 m/s².
Identify the error, correct it and name a check you can now do independently.

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: 1 2 3

07 Exit check

Two horizontal launches from the same height have speeds 2 and 6 m/s.
Compare fall times and ranges.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Two balls launch horizontally from the same platform at 2 and 5 m/s. Their shared fall time is 3 s.
Draw paths and compare ranges.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3